# Bid intelligence for Weibee Steel: the full walk-through

Where we win, what the work actually earns, and which of the next two years' opportunities deserve estimating hours. 413 bids, 36 general contractors, 90 upcoming projects, all synthetic.

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# What this deck covers

1. **The problem and the approach**
   What we cannot answer today, the four-step loop, the data it needs
2. **What the data says**
   Hit rates, pricing feedback, customers, estimate versus actual, shop capacity
3. **The decision tool**
   How the scorecard scores a project, a worked example, the ranked list, the rules
4. **How much to trust it**
   What drives the win odds and how the model tested out of sample
5. **Making it real**
   From synthetic to Weibee data, a 90-day plan, technology, risks, the monthly KPIs

# 1. The problem and the approach

Every bid teaches us something. Today the lesson stays in people's heads.

# Four questions a fabricator should be able to answer, and usually cannot



## How often do we win, by what?

Hit rate by project type, size, distance, customer and how the job is bought, with a denominator. Not a feeling about schools.



## What did the work actually earn?

Estimated versus realised gross margin by kind of job, so the next estimate learns from the last close-out.



## Why did we lose?

Price gap to the low number when we lost on price; relationship, schedule or scope when we did not. Collected every time.



## Is this invitation worth a week?

A ranked list of upcoming work, scored on odds, margin, estimating cost and shop load, before the takeoff starts.

# The loop: collect, measure, rank, decide



## 1. Collect

Five tables: customers, opportunities (bid or not), bids, jobs, upcoming pipeline. Plus the shop calendar.

*Produces: one source of truth*



## 2. Measure

Hit rate by count and by dollars. Won dollars per estimating hour. Realised margin by kind of work. Price gap on losses.

*Produces: the report's charts*



## 3. Rank

Win odds from our own history. Expected margin dollars net of estimating cost. Shop load if we win it.

*Produces: the scorecard*



## 4. Decide

Monday bid meeting works the ranked list. The room overrides, the override is logged and becomes training data.

*Produces: better odds next quarter*

# The data it needs: five tables, most of which already exist somewhere

| Table | One row per | Core fields | Where it lives today |
|---|---|---|---|
| Customers | General contractor or owner | Name, HQ, tier, first contact | Estimators' contacts, QuickBooks |
| Opportunities | Project we heard about, bid or not | GC, type, city, miles, tons, complexity, delivery method, scope, due date, fab window, bid / no-bid and why | Email, the whiteboard |
| Bids | Number we submitted | Estimating hours, cost build-up, markup, price, bidders, result, lost reason, low bid number | Estimating log, bid tabs |
| Jobs | Won bid, filled in as it runs | Actual hours and material, change orders requested and approved, final revenue, schedule variance, NCRs, days to pay | Job cost, PM close-outs |
| Pipeline | Project expected to bid in 6 to 24 months | Same as opportunities plus status: lead, invited, bidding | Does not exist yet |

**The two fields that are missing almost everywhere:** the no-bid decision with its reason, and the low number when we lose on price. Coverage of those two fields is the first KPI I would report, because every other number depends on them.

# 2. What the data says

413 bids over four and three-quarter years, 95 completed jobs, 36 general contractors. Synthetic, but shaped like a real shop.

# The business at a glance, trailing twelve months

## 25%

**Hit rate**

20 of 80 decided bids; prior year 33%

## $19.4M

**Value won**

5,088 tons

## 16 weeks

**Won work waiting for the shop**

2,286 tons at 140 tons a week

## 50%

**How full the shop looks next year**

won work plus upcoming work at its odds

## 4.5 pts

**Margin estimated but not kept**

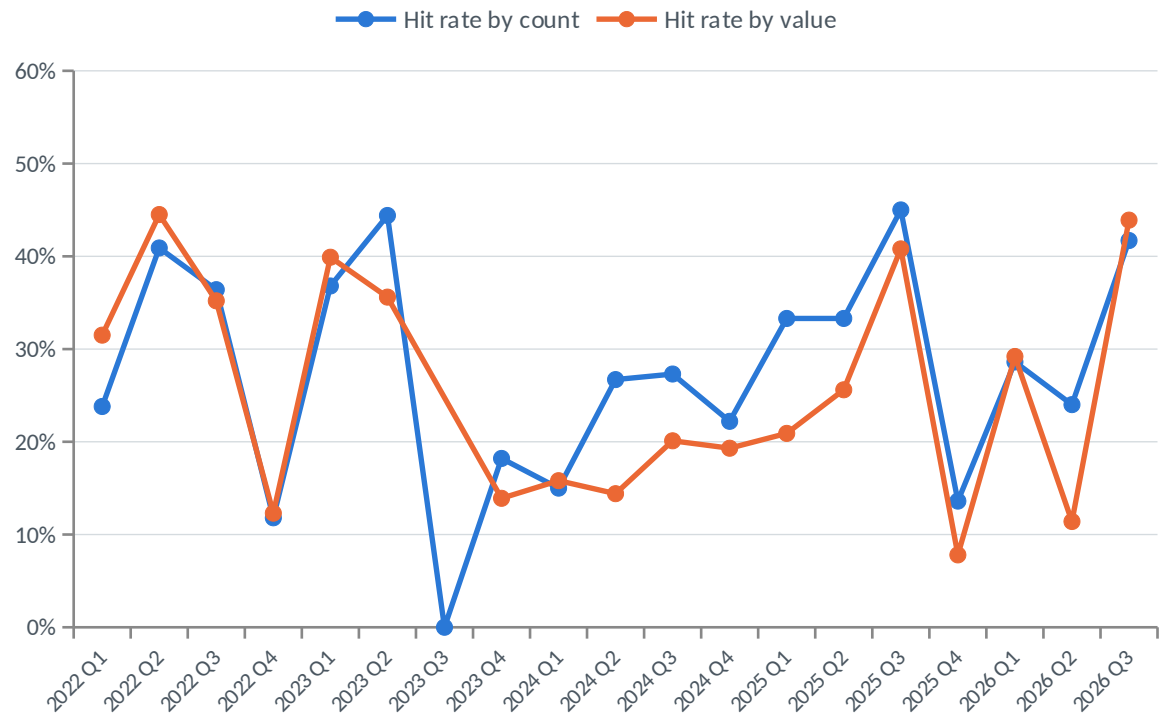
estimated 22.9%, actually earned 18.3%

## 29 of 90

**Upcoming projects worth chasing**

about 698 estimating hours to bid them

# Hit rate by quarter: by count and by dollars



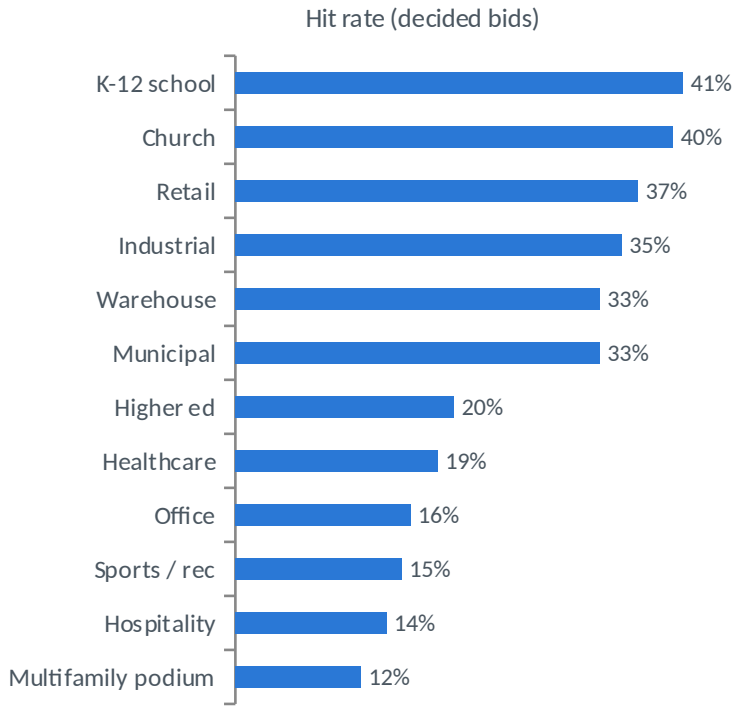

**Decided bids only; cancelled projects do not count against us.**

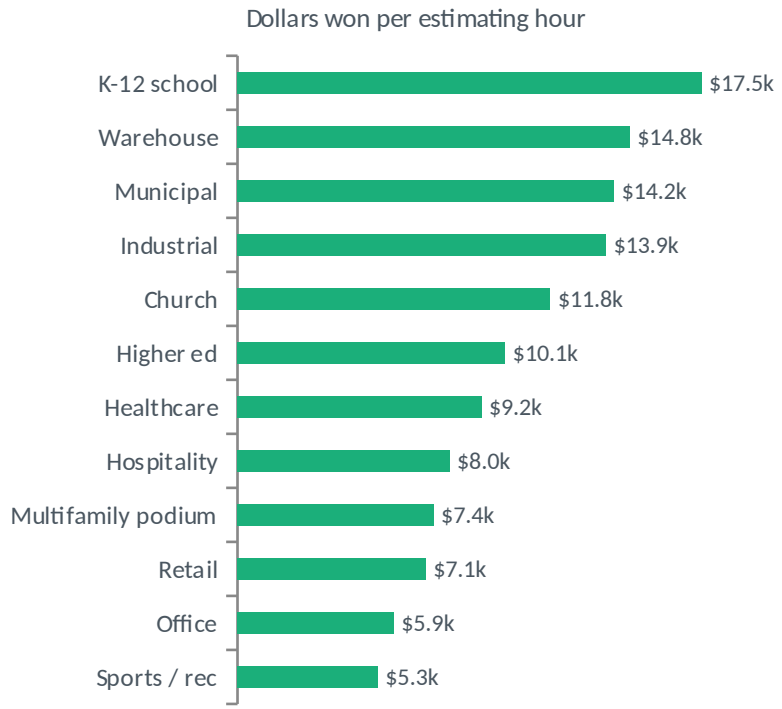
Long-run average 27%. Quarters swing because 20 bids is a small sample; one quarter is weather, four is a trend.

Count and dollar rates move together, so we are not winning only the small jobs.

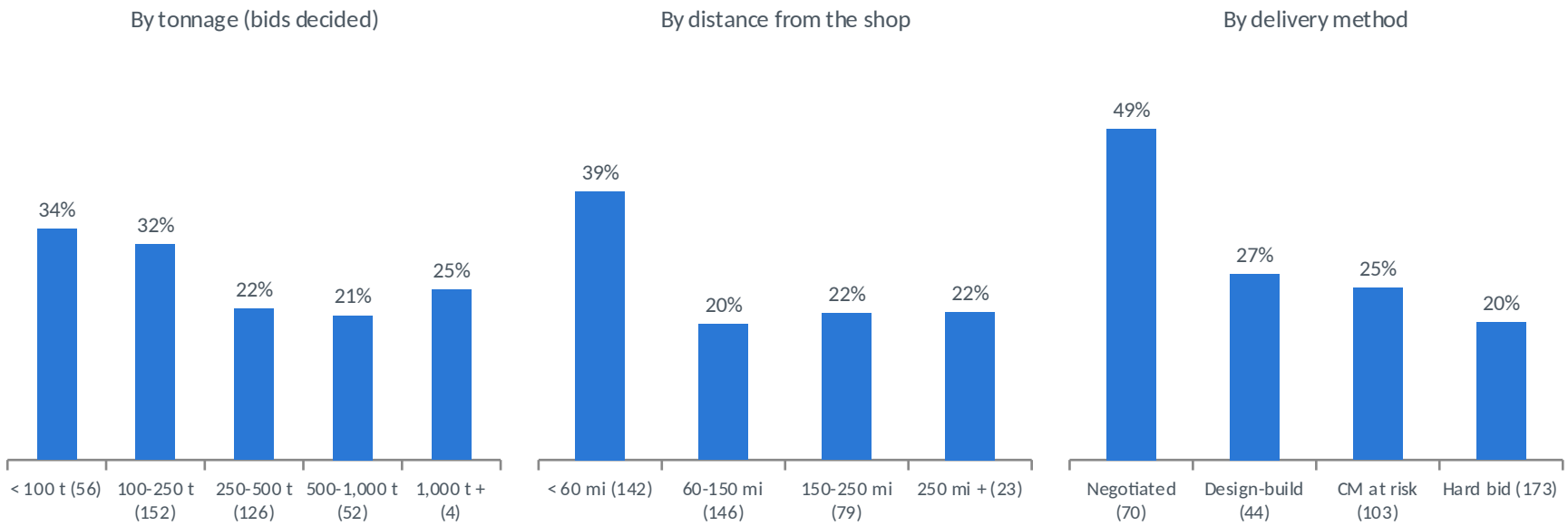
**Report both every quarter and never react to one.**

# Where the shop fits: hit rate and dollars won per estimating hour, by project type



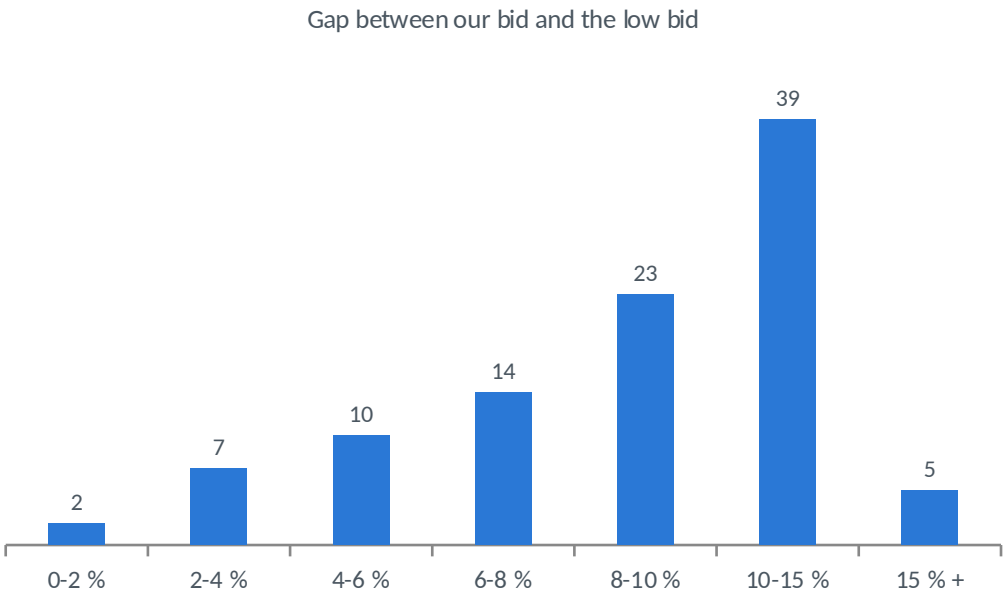



# Hit rate by size, distance and how the job is bought




**Reading it:** negotiated and design-build work wins roughly twice as often as hard bids; hit rate falls off beyond 250 miles (23 bids); and the mid-size bands carry most of the dollars. None of these cuts is a reason to never bid, all of them are a reason to bid differently.

# Pricing feedback: how far above the low number were we when we lost on price?




**9.7%**

**Median gap when we lose on price**

feedback collected on 62% of price losses

**3.9%**

**Left on the table when we win**

median gap to the next-lowest bid, where known

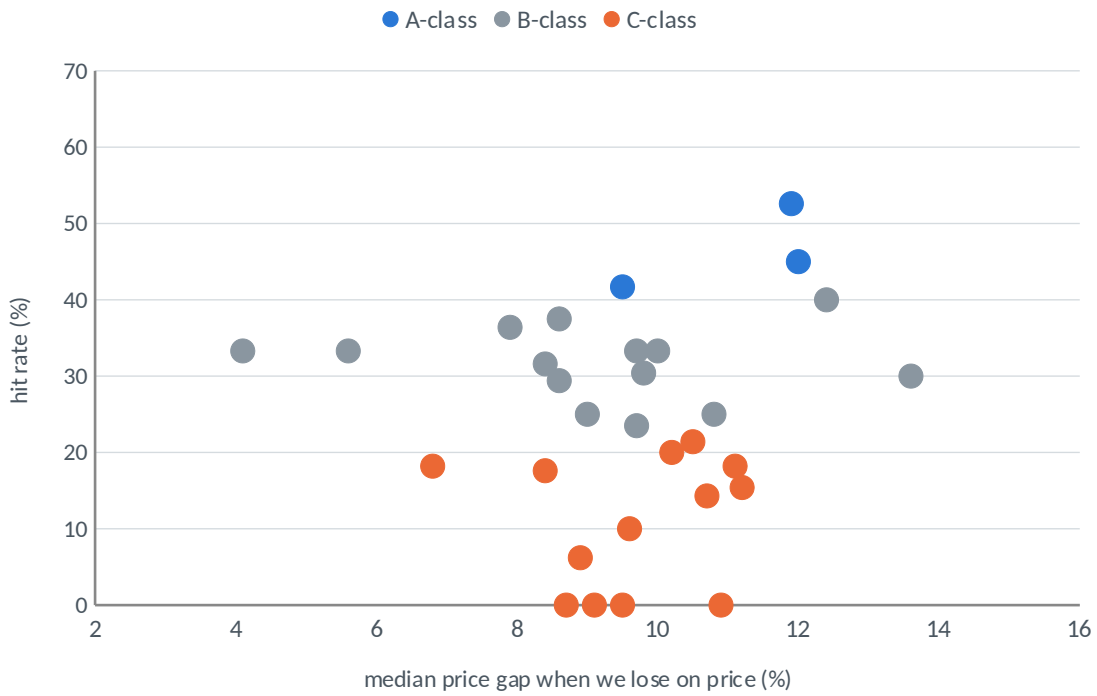
**Rule of thumb:** a gap under 3% is a markup conversation. A gap over 10% means we should not have been on that bid list, and the scorecard should learn that. Most of our losses sit in the second group.

# Customer scorecard: the ten general contractors we have won the most from

| General contractor | Tier | Class | Bids | Hit rate | Won | Won / est. hr | Price gap | Realised margin | Days to pay |
|---|---|---|---|---|---|---|---|---|---|
| Sooner State Contractors | regional | A | 20 | 45% | $11.8M | $18k | 12.0% | 17.2% | 62 |
| Granite Peak Builders | regional | A | 24 | 42% | $8.4M | $14.2k | 9.5% | 20.5% | 58 |
| Wagon Trail Builders | national | A | 19 | 53% | $8M | $16.4k | 11.9% | 22.6% | 47 |
| Ouachita Building Group | regional | B | 17 | 24% | $6.8M | $13.1k | 9.7% | 15.4% | 49 |
| Meridian Build Partners | regional | B | 23 | 30% | $6.5M | $12.2k | 9.8% | 17.7% | 61 |
| Washita Valley Builders | local | C | 17 | 18% | $6.3M | $15.4k | 8.4% | 15.8% | 63 |
| Longhorn Commercial | regional | B | 9 | 33% | $6.1M | $21.4k | 5.6% | 28.3% | 47 |
| Great Plains Constructors | national | B | 18 | 33% | $5.6M | $12.1k | 9.7% | 22.6% | 71 |
| Red River Builders | regional | B | 19 | 32% | $5.5M | $12.3k | 8.4% | 20.1% | 42 |
| Verdigris Construction | local | B | 9 | 56% | $4.8M | $19.5k |  | 15.0% | 62 |

A = awards us 40% or more and the work earns at or above shop margin. B = 22% or more. C = below. Hit rate alone flatters a GC who sends us everything; realised margin, change-order behaviour and payment speed belong next to it.

# Who awards us work and who checks their numbers against us



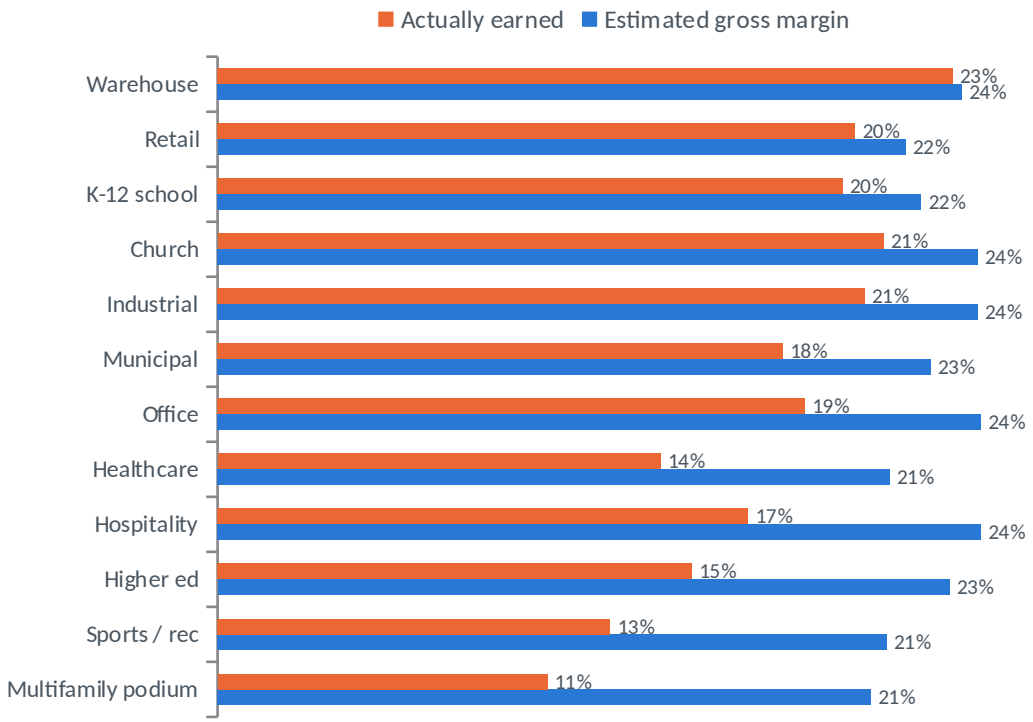

**Bottom right is the expensive corner: GCs who rarely award us and are far below us on price.**

Top left is where the business is: GCs who award us often and whose price gap is small when they do not.

Each dot is a GC with three or more decided bids.

**Spend estimating hours top-left, and on new GCs that look like them.**

# What the estimate promised versus what the job earned, by project type



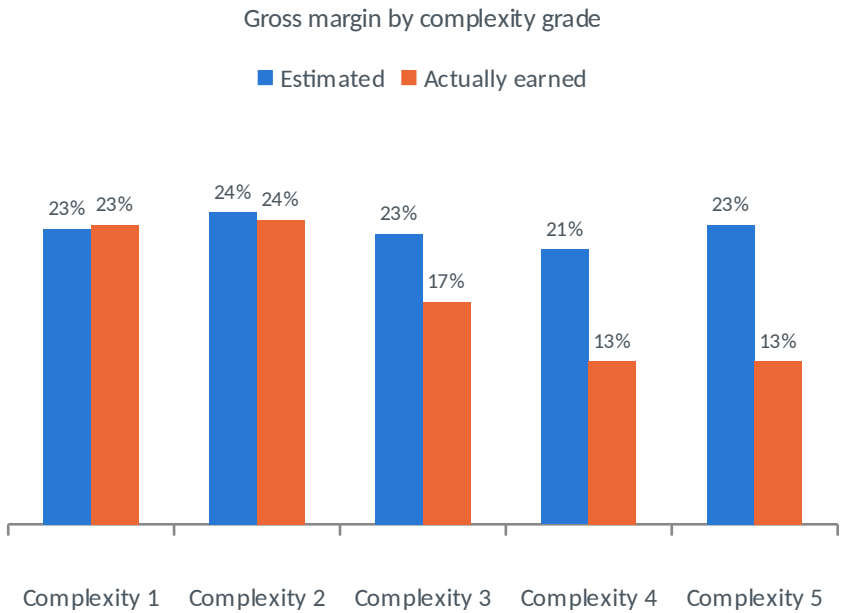

**Across 95 completed jobs: 22.9% estimated, 18.3% earned.**

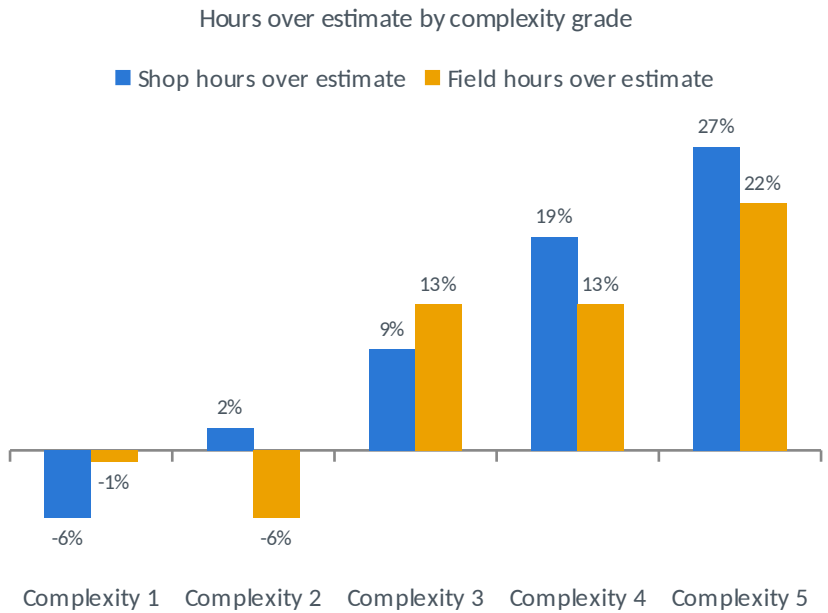
4.5 points, about $4.3M, that was in the bid and not in the bank.

Sorted worst to best. The estimate is roughly right on simple work and roughly four to ten points optimistic on complex work.

**Put realised margin by type into the estimate as a line item.**

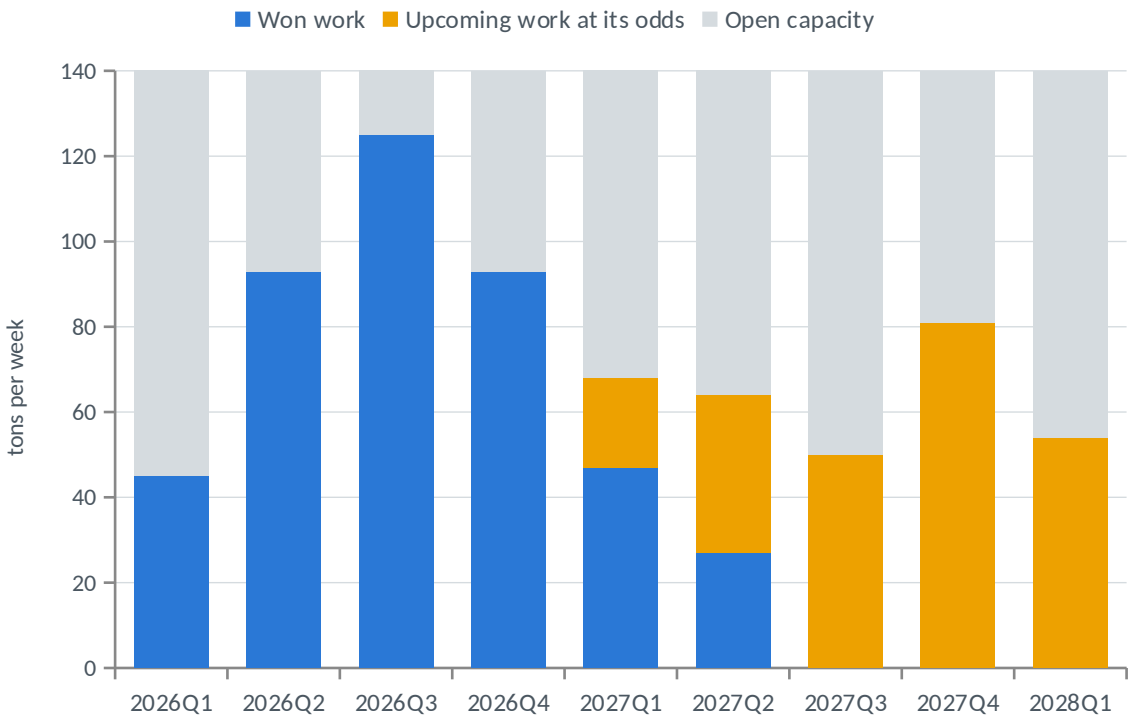
# Complexity is where the margin goes: hour overruns by complexity grade




**Grades:** 1 is simple braced frames, 5 is architecturally exposed steel, long spans or heavy moment connections. Shop hours ran 10% over estimate overall and field hours 8%; the overrun grows with the grade. A complexity-based contingency, not a flat markup, is the fix.

# Can the shop take what we are chasing? Backlog and weighted pipeline against capacity




**Each won job spreads its tonnage evenly over its fabrication weeks.**

Each upcoming job counts at tons times win odds: a 400-ton job at 25% is 100 tons.

Won work covers 16 weeks. From early next year the shop is less than half full unless the pipeline converts.

**Target: roughly 9,000 more tons of identified opportunities at today's hit rate.**

# 3. The decision tool

Every upcoming project scored before a week of takeoff starts. A starting point for the bid meeting, not a verdict.

# How the scorecard scores a project

### Win odds

From a model fitted to our own bids: project type, tonnage, miles, complexity, delivery method, bidders, and our history with that GC.

### Expected revenue

Tons times our historical price per ton for that project type (the real estimate replaces this once one exists).

### Margin we really keep

Realised gross margin on that project type from completed jobs, not the estimate's margin.

### Estimating cost

Hours from a curve fitted to our own bid log (tons and complexity) times the loaded estimator rate.

**Expected value = win odds x expected revenue x margin we really keep minus estimating cost**

Ranked per estimating hour, because hours are the scarce input, and compared with what an hour has returned historically (about $2,468 of gross margin per estimating hour over the last two years). Pursue at 80% of that or better, Review from 35% to 80%, Pass below. A separate check projects shop load over the job's fabrication weeks if we win it.

# Worked example: one Pursue and one Pass, step by step

## Pursue Industrial - Stillwater #24

Caddo Construction, Stillwater, 83 mi, 398 t, complexity 1, Design-build, 3 prior bids with this GC at 14% hit rate

| Step | Value |
|---|---|
| Win odds | 52% |
| Expected revenue | 398 t x $4,011 = $1.60M |
| Margin we really keep | 19.9% |
| Expected margin dollars | 52% x $1.60M x 19.9% = $167k |
| Estimating cost | 25.8 h x $85 = $2k |
| Expected value | $164k |
| EV per estimating hour | $6,373 (258% of historical) |
| Shop load if won | 82% |
| Reasons | design-build delivery |

## Pass Warehouse - Fort Smith #83

Longhorn Commercial, Fort Smith, 195 mi, 711 t, complexity 1, Hard bid, 9 prior bids with this GC at 32% hit rate

| Step | Value |
|---|---|
| Win odds | 11% |
| Expected revenue | 711 t x $3,130 = $2.23M |
| Margin we really keep | 21.9% |
| Expected margin dollars | 11% x $2.23M x 21.9% = $51k |
| Estimating cost | 37.4 h x $85 = $3k |
| Expected value | $48k |
| EV per estimating hour | $1,283 (52% of historical) |
| Shop load if won | 79% |
| Reasons | win odds 11% |

# The ranked pipeline: 90 upcoming projects on one chart




**Pursue** — **29 projects**

698 estimating hours
9,872 tons, $2.5M expected margin net of estimating

**Review** — **26 projects**

451 estimating hours
4,608 tons, $0.72M expected margin net of estimating

**Pass** — **35 projects**

858 estimating hours
9,961 tons, $0.53M expected margin net of estimating

# The top ten, as the bid meeting would see them

| # | Call | Project | GC | Tons | Miles | Delivery | Odds | EV / hr | Shop load | Why |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Pursue | Industrial - Stillwater #24 | Caddo Construction | 398 | 83 | Design-build | 52% | $6.4k | 82% | design-build delivery |
| 2 | Pursue | Municipal - Norman #23 | Flint Hills Construction | 134 | 13 | Negotiated | 83% | $5.6k | 74% | negotiated delivery |
| 3 | Pursue | Industrial - Muskogee #3 | Meridian Build Partners | 494 | 133 | Design-build | 43% | $5.2k | 72% | design-build delivery |
| 4 | Pursue | Industrial - Oklahoma City #11 | Boomer Construction Group | 923 | 32 | CM at risk | 36% | $5k | 69% | no flags |
| 5 | Pursue | K-12 school - Tulsa #26 | Sooner State Contractors | 175 | 120 | Negotiated | 65% | $4.7k | 63% | relationship GC (43% hit rate, 20 bids); project type we w… |
| 6 | Pursue | Industrial - Oklahoma City #45 | Summit Ridge Construction | 1330 | 30 | Hard bid | 24% | $4.2k | 69% | no flags |
| 7 | Pursue | Municipal - Tulsa #61 | Meridian Build Partners | 196 | 126 | Design-build | 53% | $4.1k | 62% | design-build delivery |
| 8 | Review | Municipal - Muskogee #32 | Cimarron Commercial | 145 | 136 | Negotiated | 62% | $3.9k | 98% | shop tight (98%) during fabrication window; negotiated del… |
| 9 | Pursue | Industrial - Stillwater #49 | Flint Hills Construction | 338 | 83 | CM at risk | 37% | $3.9k | 55% | no flags |
| 10 | Pursue | Industrial - Moore #30 | Washita Valley Builders | 414 | 21 | CM at risk | 36% | $3.7k | 75% | no flags |

*Full list of 90 is in the report and the CSV, with every input next to every call so the room can argue with the number.*

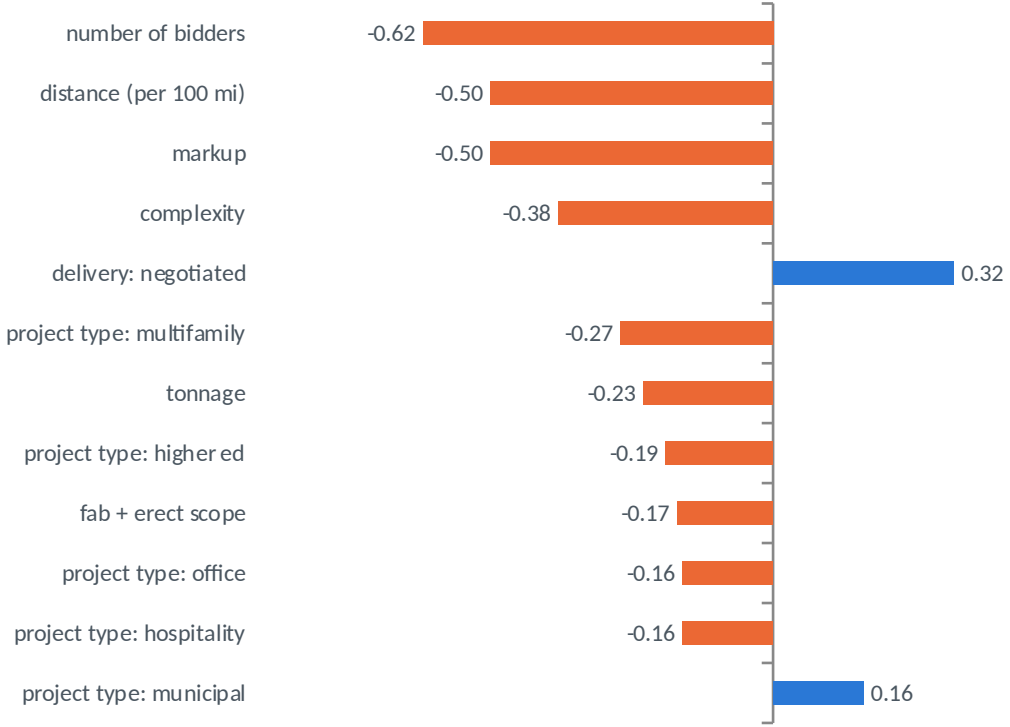
# The rules, in the order they fire, and how the room overrides them

| | | |
|---|---|---|
| 1 | **Hard flags first** | Shop over 105% of capacity if we win it, or win odds under 12%, is a Pass whatever the dollars say. |
| 2 | **Then the money** | Expected value per estimating hour against our historical return: 80% or better is Pursue, 35% to 80% is Review, below is Pass. |
| 3 | **Then the soft flags** | Shop tight (over 90%), erection beyond 250 miles, a project type where we realise well below shop margin: a would-be Pursue becomes Review. |
| 4 | **One strategic override** | A new regional or national GC on one of our three best-fit project types is never an automatic Pass; it is Review, labelled strategic. |
| 5 | **The room decides** | Every call shows its inputs and reasons. When the room overrides, the override and the reason are logged and become next quarter's training data. |

# 4. How much to trust the win odds

A small statistical model on our own bids, tested on bids it had never seen. Good enough to rank, not good enough to overrule the room on one job.

# What moves the win odds, in order of effect



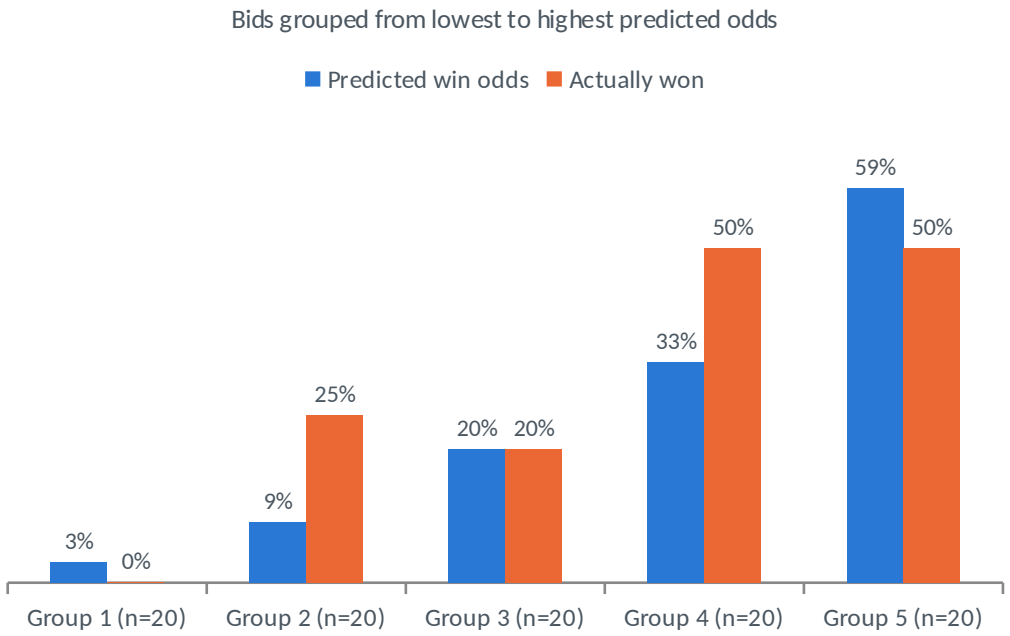

**Bars to the left lower our odds, bars to the right raise them.**

More bidders, more miles, a higher markup and higher complexity all cost us. Negotiated delivery and a good history with the GC help.

Measured against warehouse work on a hard bid.

**Nothing here surprises an estimator. That is the point: the model agrees with experience and adds a number.**

# Did the odds come true? Tested on bids the model had never seen




**0.75**

**Accuracy score (AUC)**
0.5 is a coin flip, 1.0 is perfect

**0.182 vs 0.206**

**Error score vs guessing the average**
Brier score, lower is better

**100 bids**

**Held out for the test**
fitted on 290 bids before 2025-07-01

**How to use it:** the groups line up, so a 40% call wins about 40% of the time. Use the odds to rank and to budget hours. Do not use them to skip a job the room believes in. Re-fit every quarter; every decided bid and every low-bid number makes it better.

# 5. Making it real at Weibee Steel

What changes with real data, the first 90 days, the technology worth evaluating, the risks, and what we would report every month.

# From the sample to Weibee's own numbers

| Piece | In this sample | On real data |
|---|---|---|
| Source data | Generated to behave like a mid-size fabricator | Estimating log, job-cost close-outs, GC list, loaded through the same five tables |
| Capacity | 140 tons a week, one step change | The shop calendar with shifts, holidays and outsourced fabrication |
| Expected revenue | Historical price per ton by project type | The actual estimate once one exists; type median only at lead stage |
| Win model | Type, size, miles, complexity, delivery, bidders, GC history | Add the estimator, bid-tab bidder names, time since last award, contact changes |
| Thresholds | Relative to our historical return per estimating hour | Same; reviewed quarterly with the estimating manager |
| The report | One HTML page, a PDF, a Tableau data layer | Same files on the shared drive, or the same tables behind Power BI or Tableau |
| Findings | Illustrate the kind of result | Become decisions with names and dates attached |

# The first 90 days

**Weeks 1 to 4**

## Learn the shop, build the intake

- Shadow estimating, PM, QC, shop and field as the posting describes
- Stand up the five tables; load three years of bid logs and job closes
- Phone the GCs we lost to for the low number; log no-bids going forward
- KPI: coverage of lost reason and low-bid fields

**Weeks 5 to 8**

## Get the numbers live

- Hit rate, customer and estimate-versus-actual pages on real history
- Sit with estimators and PMs until definitions are agreed: decided bid, direct cost, change-order booking
- First account plan from the customer scorecard
- KPI: the report runs weekly without me touching it

**Weeks 9 to 13**

## Turn on the Monday scorecard

- Score the pipeline; set the quarterly estimating-hour budget
- Log every override with a reason
- Monthly capacity review with shop leadership
- KPI: share of estimating hours on Pursue work; pipeline tons versus the gap

# Technology worth evaluating, in this order, and how we would judge each



### 1. GC and opportunity database

A spreadsheet with discipline first, then a light CRM. Judged on whether the intake survives staff changes.



### 2. Automated and AI-assisted takeoff

Model-based quantities from Revit or IFC, or AI-assisted PDF takeoff. Judged against our fitted hours-per-bid curve on the project types we actually pursue.



### 3. Lead services

Dodge, ConstructConnect, BuildCentral. Every lead runs through the scorecard, so we buy leads where the model says we win.



### 4. Pricing feedback

A post-award email template and a field in the database before any software. Cheapest, highest return.

# Risks and limits, stated up front

### Garbage in

If lost reasons and low-bid numbers are not collected, the pricing section and the odds degrade quietly. Mitigation: coverage is a reported KPI.

### Small samples

A few hundred bids. Quarterly hit rates swing, project-type effects are noisy. Mitigation: report sample sizes, shrink thin cuts toward the average.

### False precision

A 38% win probability is not a fact. Mitigation: show odds as ranges in the room, never overrule a person on one job.

### Gaming

If hours or markup are judged by the scorecard, people will manage to it. Mitigation: the scorecard allocates hours; it does not grade estimators.

### Scope

Only estimating hours are modelled as scarce. Bonding, cash, crews and GC schedule risk belong in the reasons column once data exists.

### This sample

All data is synthetic. Findings illustrate the kind of result, not Weibee Steel.

# What we would report every month

| KPI | Definition | Why it matters |
|---|---|---|
| Hit rate, count and value | Wins over decided bids; dollars won over dollars bid, trailing 12 months | The headline, both ways, so we see if we are winning only the small jobs |
| Won dollars per estimating hour | Value won over all estimating hours spent, by project type | The return on the estimating team's time |
| Feedback coverage | Share of price losses with the low number recorded; share of no-bids with a reason | Everything else depends on it |
| Median price gap | Our bid over the low bid when we lost on price, by delivery method | Markup problem or fit problem |
| Margin slip | Estimated minus realised gross margin on jobs closed this quarter, by type and complexity | Whether the estimate is learning |
| Pipeline coverage | Backlog plus weighted pipeline over capacity, next four quarters | The business-development target in tons |
| Hours on Pursue work | Share of estimating hours spent on Pursue calls; override count and reasons | Whether the scorecard is being used, and where it is wrong |
| A-class share | Share of bids and of won dollars from A-class accounts; new A-class candidates | Account planning |

# Every bid teaches us something. This puts the lesson in a table and on the Monday agenda.

Report, PDF, Tableau data, technical design and code are in the repository. All data is synthetic; the method runs unchanged on real history.

Thank you